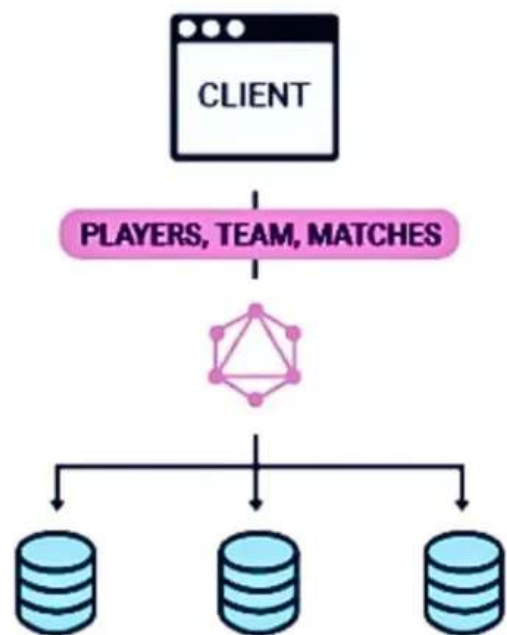
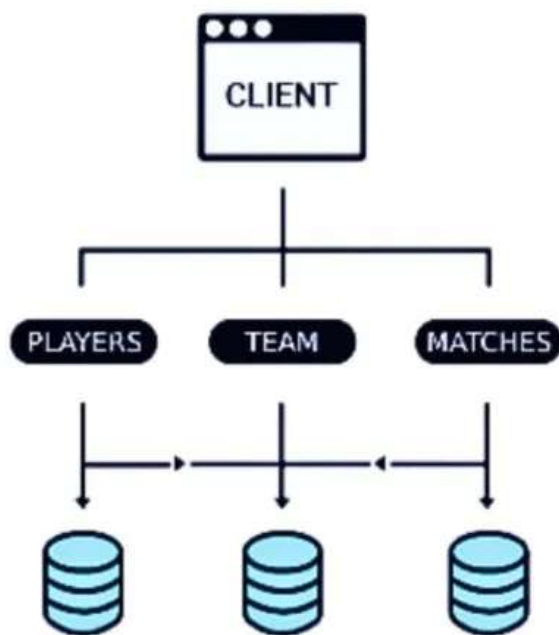


GraphQL vs REST API

(With Examples)



Swipe>>

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Data Fetching

REST: Uses **fixed endpoints** (/users, /posts) to retrieve data.

Request GET /users/1

Response { "id": 1, "name": "John", "email": "john@email.com" }

GraphQL: Allows querying specific data fields in a single request.

```
{
  user(id: 1) {
    name
    email
  }
}
```

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GraphQL Client

```
{  
  assets (<album_id>) {  
    id,  
    url,  
    comments {  
      text  
    }  
  }  
}
```

```
{  
  assets: [  
    { id: 1,  
      url: '...',  
      comments: [  
        { text: '...' }  
      ]  
    },  
    { id: 2,  
      url: '...',  
      comments: [  
        { text: '...' }  
      ]  
    }  
  ]  
}
```



GraphQL Server

Coding Notes in bio

REST Client

GET /albums/: album_id/assets

```
{  
  data: [  
    { id: 1, url: '...' },  
    { id: 1, url: '...' }  
  ]  
}
```

GET /assets/:asset_id/comments
(for each asset)

```
{  
  data: [  
    { author_id: 32, url: '...' },  
    { author_id: 243, url: '...' }  
  ]  
}
```



REST Server

Efficiency

REST: Often **requires multiple requests** for related data.

```
GET /users/1
```

```
GET /users/1/posts
```

GraphQL: Fetches all necessary data in a **single request**.

```
{  
  user(id: 1) { name email posts { title content } }  
}
```

Response Format

REST: Returns a predefined JSON structure.

```
{
  "id": 1,
  "name": "John",
  "email": "john@email.com",
  "age": 30,
  "posts": [
    { "id": 101, "title": "GraphQL Basics", "content": "..." }
  ]
}
```

GraphQL: Returns only the requested data & eliminates unnecessary fields, optimizing responses.

```
{
  "user": {
    "name": "John",
    "email": "john@email.com"
  }
}
```

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Use Cases

REST API: Best for **simple, cache-friendly APIs** with fixed structures.

Use REST when: Public APIs & caching are needed.

GraphQL: Best for **complex, dynamic apps needing flexible data fetching.**

Use GraphQL when: Frontend needs customized queries.